SymAlign-UAST: Neuro-Symbolic Code Understanding via Hyperbolic Geometry and Structural Invariance

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Code structure remains informative when identifiers fail.

100% Obfuscation Retention
- Identifier-invariant by construction

0.680 → 0.803
- Class separation with hyperbolic retrieval

+0.04 → −0.33
- Entropy–violation alignment

Identifiers removed. Structure preserved.

```python
import qiskit

def build_three_qubit_circuit():
    qc = QuantumCircuit(3)
    qc.ccx(0, 1, 2)
    qc.cx(0, 2)
    return qc
```

```python
import <MASKED>

def <MASKED>():
    <MASKED> = <MASKED>(<MASKED>)
    <MASKED>.<MASKED>(...)
    <MASKED>.<MASKED>(...)
    return <MASKED>
```

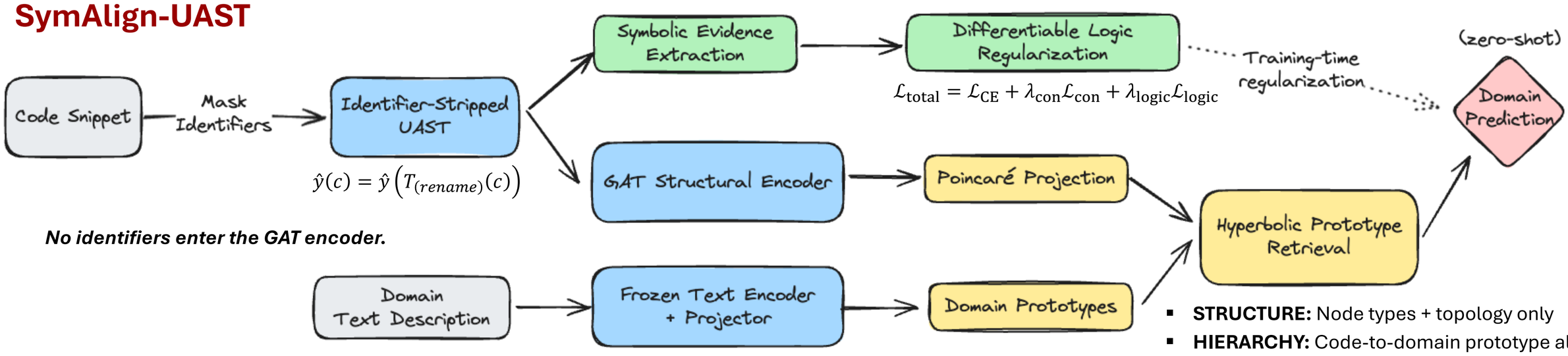
```
UAST Graph

Root
  ├─ Package
  └─ Function
      ├─ Assignment
      ├─ Call
      ├─ Call
      └─ Return
```

Why Structure?

When identifiers fail, can code structure still reveal domain?
- Obfuscated Code
- Stripped Binaries
- Malware Triage
- Minified Code
- Analyst Routing
- Cross-Language Transfer

SymAlign-UAST

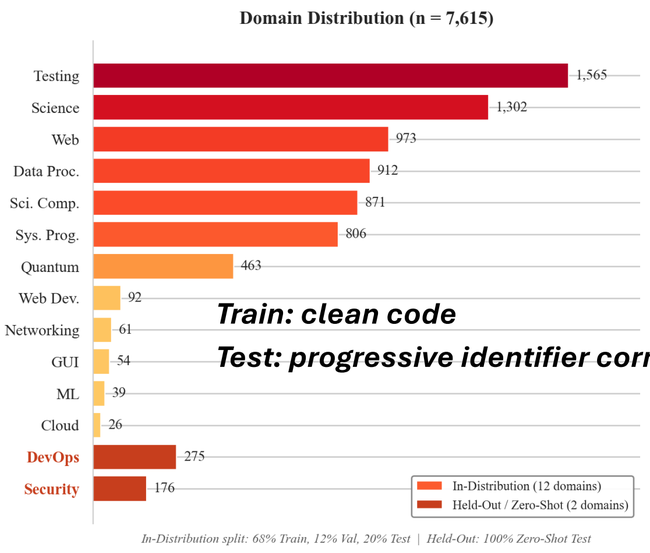


No identifiers enter the GAT encoder.

- **STRUCTURE:** Node types + topology only
- **HIERARCHY:** Code-to-domain prototype alignment
- **LOGIC:** Evidence-guided confidence calibration

Experimental Setting

7,615 Python files · 247 repositories · 14 domains
- Security and DevOps: zero-shot nearest-prototype retrieval.



Train: clean code

Test: progressive identifier corruption

In-Distribution split: 68% Train, 12% Val, 20% Test | Held-Out: 100% Zero-Shot Test

Robustness under Obfuscation

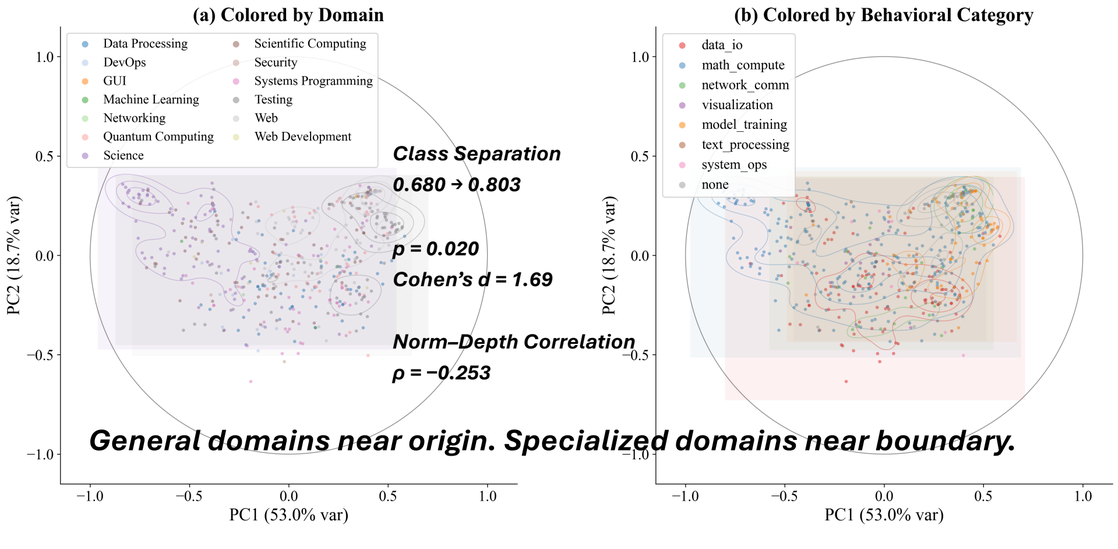
Surface-form accuracy depends on import identifiers.
- Perfect retention under full identifier stripping.

- GraphCodeBERT: 96.59% → 13.89%
- TF-IDF + LR: 93.62% → 12.63%

Model	Clean (C0)	Import-Masked (C1)	Full Obfusc. (C4)	Retention
GraphCodeBERT	**96.59**	19.21	13.89	0.144
TF-IDF + LR	93.62	12.63	12.63	0.135
GNN-only (structure)	47.79	47.79	47.79	**1.000**
SymAlign-UAST (ours)	31.91	31.91	31.91	**1.000**

Hierarchy-Aware Retrieval

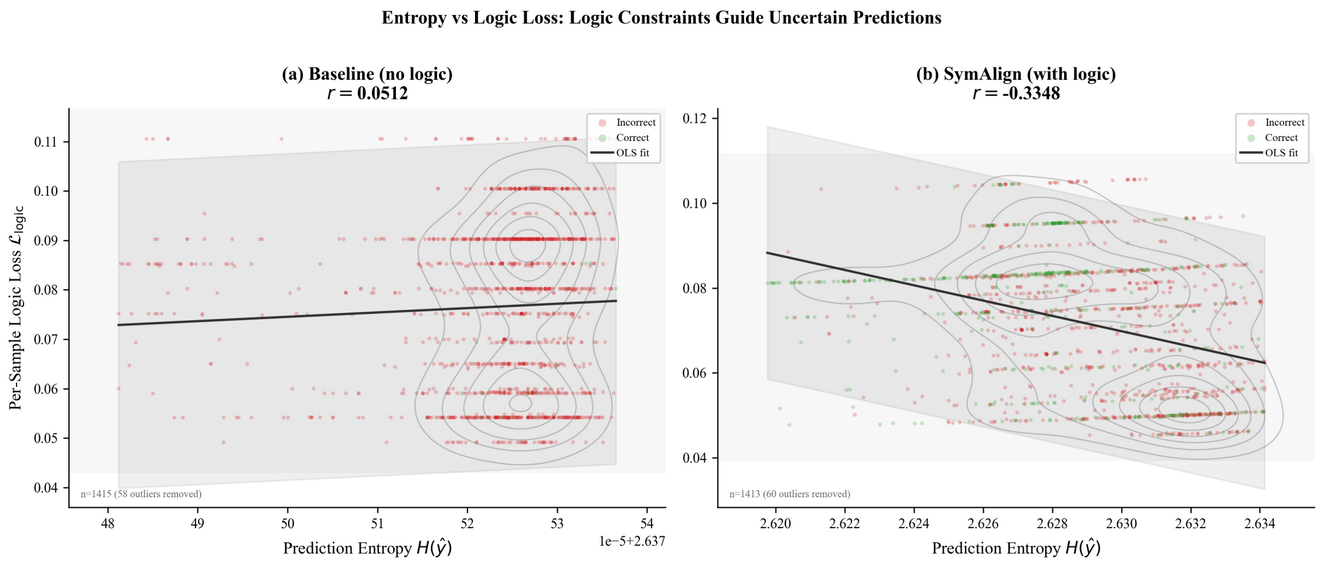
Hyperbolic space organizes domain hierarchy.



General domains near origin. Specialized domains near boundary.

Logic-Aware Reliability

Soft constraints preserve accuracy better than hard correction.



- Higher inconsistency ----------> Higher uncertainty
- Modest aggregate gain, directional reliability signal

Identifiers are brittle. Structure is invariant. Structure-first code understanding under identifier corruption.